Length Is the Floor: A Length-Residualized Audit of Training-Free Dynamic-RAG Triggers

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Abstract

Dynamic retrieval-augmented generation systems decide *when* to retrieve from a scalar trigger signal, scored by raw AUROC against binary correctness. Recent work has shown this protocol is length-confounded on adjacent question-answering (QA) selective-prediction and uncertainty-quantification (UQ) tasks. We present the first length-residualized audit of the training-free dynamic-RAG trigger pool on a paired-oracle target (`lift_em`$=+1$) that asks whether retrieval flips a wrong answer to a right one, isolating the causal retrieval-helpfulness signal from length-correlated incorrectness. Across two open-weight chat models on HotpotQA and 2WikiMultiHopQA (n=800 paired traces), nine signals beat the length baseline at 95% paired-bootstrap and only four survive Bonferroni at k=15. On the held-out 2Wiki cells, raw length collapses to at-or-below chance (AUROC ≤ 0.52), and our **FCT** (first-commitment time, a forward-looking answer-trajectory signal) reaches raw AUROC **0.784** on Gemma $\times$ 2Wiki, the strongest length-orthogonal signal in the pool. The survivor set is robust to a GPT-4o-Mini semantic-equivalence judge ($\kappa = 0.707$), and FCT itself *strengthens* from $\Delta = +0.190$ to $\Delta = +0.222$ under the judge, the opposite of what a string-level (exact-match) EM artifact would predict.

1 Introduction

Dynamic retrieval-augmented generation (RAG) systems retrieve external evidence during decoding to ground generation in source material (Lewis et al., 2021; Trivedi et al., 2023; Shao et al., 2023). At the core of every such system sits a *retrieval trigger*: a scalar function of the partial generation that decides when to fire retrieval, evaluated by ranking traces by the trigger score and reporting raw AUROC against a binary correctness label. We restrict our audit to *training-free* triggers. Prior work

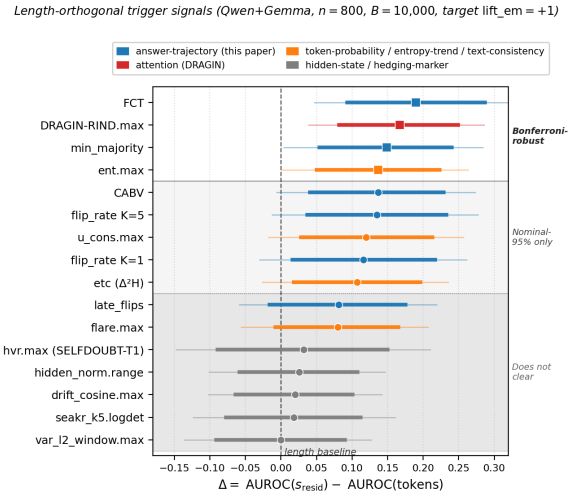


Figure 1: **Four trigger signals from three families carry length-orthogonal information beyond the length baseline; FCT (this paper) is strongest.** Forest plot of $\Delta = \mathrm{AUROC}(s_{\mathrm{resid}}) - \mathrm{AUROC}(\texttt{tokens_generated})$ on `lift_em`$=+1$ (n=800 paired traces). Thick bars: 95% paired-bootstrap CIs; thin extensions: Bonferroni-corrected 99.67% CIs at k=15. Setup: §3. Tables: App. A, B.

spans six signal families: attention-based DRAGIN (Su et al., 2024), token-probability FLARE (Jiang et al., 2023), entropy-trend ETC (Li et al., 2026), hidden-state SeaKR (Yao et al., 2025) and INSIDE (Chen et al., 2024), multi-sample text-consistency RSUS (Guo et al., 2026), and hedging-marker SELFDOUBT-T1 (Pandey et al., 2026). A contemporaneous answer-trajectory family tracks step-level dynamics of the model's own predicted answer (Datta et al., 2026; Wang et al., 2025; Chen et al., 2026).

Recent work has shown this evaluation convention is length-confounded on adjacent QA selective-prediction and uncertainty-quantification (UQ) tasks. Santilli et al. (2025) prove that mutual length-bias on UQ scores and correctness functions systematically distorts AUROC. Janiak et al. (2025)

show the empirical consequence: detection AUROC drops by up to 45.9% when switching from ROUGE to an LLM-as-Judge label. Zhang et al. (2024) attribute the underlying behavior to verbosity compensation on multi-hop QA, and Ielanskyi et al. (2025) document a related correctness-function failure across approximate metrics. The methodological response is Vashurin et al. (2025)'s UNCERTAINTY-LINE, which regresses each UQ score on output length and uses the residual as a length-invariant estimate. The dynamic-RAG trigger pool is the missing application.

We extend the length-residualization protocol of Vashurin et al. (2025) to the training-free dynamic-RAG trigger pool. Rather than score triggers against single-arm correctness, we re-decode each closed-book trace under gold context and ask whether retrieval flips a wrong answer to a right one, defining a paired-oracle target $\texttt{lift_em}=+1$ conceptually motivated by Mallen et al. (2023). This isolates the *causal* retrieval-helpfulness signal from the length-correlated incorrectness signal that the published pool actually measures (§5.1). We also propose **FCT** (first-commitment time), a forward-looking variant of the answer-trajectory primitive (Datta et al., 2026; Wang et al., 2025; Chen et al., 2026; Zhang et al., 2026b). FCT reads $K=5$ majority answers from non-thinking-mode chat models, where neither Datta's `</think>`-anchored forced completion nor Zhang et al.'s finite-set continuation projection applies directly (App. E).

Contributions. (1) A length-residualized audit of the dynamic-RAG trigger pool, the first to our knowledge, scored against a paired-oracle target ($\texttt{lift_em}=+1$). (2) **FCT** (first-commitment time), our forward-looking variant of the answer-trajectory primitive and the strongest length-orthogonal trigger in the audited pool. (3) A correctness-function robustness check against Janiak et al. (2025) and Ielanskyi et al. (2025), under which the survivor matrix is preserved and FCT *strengthens*.

2 Related Work

The length-confound critique of standard UQ AUROC evaluation comes from QA UQ and hallucination detection (Santilli et al., 2025; Janiak et al., 2025; Zhang et al., 2024; Ielanskyi et al., 2025; Vashurin et al., 2025; Mao et al., 2026). The answer-trajectory family (Datta et al., 2026; Wang et al., 2025; Chen et al., 2026) provides the parent primitives for FCT. Two related supervised methods sit outside our training-free scope: Lu et al. (2026) train hidden-state probes for streaming hallucination detection, and Zhang et al. (2026a) contrastively shape answer embeddings under latent perturbation at the trace boundary, where their "answer agreement" differs from our $K=5$ boundary-completion "answer-trajectory". Trained adaptive-retrieval policies likewise fall outside our audit (Asai et al., 2023; Guo et al., 2025). Our paired-oracle target operationalizes the question of when retrieval helps a parametric model (Mallen et al., 2023), replacing static entity-popularity proxies with per-trace closed-vs-oracle re-decode.

3 Methods

3.1 Paired-Oracle `lift_em` Protocol

The published trigger pool scores against single-arm correctness, a target length predicts moderately well on multi-hop QA (§5.1). We instead define a paired target that flags traces where retrieval would convert a wrong answer to right, isolating the *causal* retrieval-helpfulness signal from length-correlated incorrectness. Concretely, for each question q, we run two greedy decodes on the same chat model: a closed-book pass that produces trace $\tau_c^{(q)}$ and answer $a_c^{(q)}$, and an oracle pass conditioned on the gold supporting paragraphs that produces $a_o^{(q)}$. We read all trigger signals from $\tau_c^{(q)}$; the oracle decode only supplies the label. The discrimination target is

$$\texttt{lift_em}(q) = \mathbb{1}\left[\mathrm{EM}(a_o^{(q)}) > \mathrm{EM}(a_c^{(q)})\right] \quad (1)$$

We evaluate on HotpotQA (Yang et al., 2018) and 2WikiMultiHopQA (Ho et al., 2020), holding the latter out as a cross-dataset cell, at $n=200$ paired traces per architecture-by-dataset cell. The two models are Qwen2.5-7B-Instruct (dense) and Gemma-4-26B-A4B-it (MoE-hybrid), chosen for cross-architecture coverage. The positive class is rare: 108 of 800 paired traces, base rate 13.5%.

3.2 Length Residualization

We adopt the residualization protocol of Vashurin et al. (2025). For each trigger signal s on a cohort of paired traces, we fit OLS of s on $\log(\texttt{tokens}+1)$ and evaluate AUROC on the residual:

$$s_{\text{resid}} = s - \hat{\alpha} - \hat{\beta}\,\log(\texttt{tokens}+1). \quad (2)$$

Intuitively, s_{resid} keeps only the part of s that length cannot linearly explain; its AUROC therefore measures whether the signal carries information beyond length. Our $\log(\texttt{tokens}+1)$ parameterization agrees within $\varepsilon \leq 0.029$ AUROC with the public reference implementation on our cohorts (App. D). Several signals in the audited pool (ETC, FLARE, SeaKR, SELFDOUBT-T1) are originally online or per-token, and we evaluate each as a trace-level summary statistic with granularity adapters in Appendix I.

3.3 FCT and the Answer-Trajectory Family

At each reasoning-step boundary (detected by tokenizer-level sentence-end token matchings), we sample K=5 boundary completions at temperature τ=0.6, top-p 0.9, and extract a per-step answer from each via a regex over the model's `Answer:` prefix. From the resulting per-step K=5 majority sequence $\mathcal{M} = (m_1, \ldots, m_n)$ we compute four trace-level aggregations:

- `flip_rate`: fraction of consecutive-step majority disagreements;
- `late_flips`: count of K=5 answer-*set* disagreements in the last half of the trace;
- `min_majority`: $1 - \min_t(\max_{\text{count}}/K)$, the worst within-step disagreement;
- **FCT**, a forward-looking first-commitment time

$$\text{FCT} = \frac{\min\{i : m_j = a_{\text{final}}\ \forall\, j \geq i\}}{\max(1, n-1)} \in [0,1]. \tag{3}$$

Intuitively, FCT is the (normalized) decode position at which the model first committed to its final answer. For example, on a 5-step trace with per-step majorities (X, X, Y, Y, Y) and final answer Y first reached at step i=3, FCT = $3/4 = 0.75$. Low FCT denotes early commitment; high FCT denotes a trace that kept flipping until close to the end of the decode, the traces where retrieval is most likely to help. All four aggregations adapt the `AnswerSwitches` primitive of Datta et al. (2026) to non-thinking-mode chat models with K=5 majority-answer extracts. Neither Datta's `</think>`-anchored forced completion nor Zhang et al.'s finite-set continuation projection applies directly in this regime (App. E).

Continuous variant. We additionally evaluate **CABV** (continuous answer-belief velocity), a between-step generalization of `flip_rate` defined in App. F and reported in Tables 2 and 3.

4 Results

4.1 Cross-Architecture Survivor Matrix

Of the 16 audited signals, nine clear the length baseline at nominal 95% paired-bootstrap, and only four survive Bonferroni correction at k=15 (Figure 1): attention-based DRAGIN-RIND ($\Delta = +0.167$), token-probability `ent.max` ($+0.137$), answer-trajectory `min_majority` ($+0.149$), and **FCT** ($+0.190$, the strongest length-orthogonal trigger in the pool). The four survivors span three of the seven declared families, which suggests the length-orthogonal signal is not a quirk of any one mechanism. Two families have no nominal-95% survivor: hidden-state (SeaKR, INSIDE, plus three geometry variants) and hedging-marker (SELFDOUBT-T1) are absorbed by length (App. G). The full table is in App. B.

4.2 Length Is Anti-Predictive on the Held-Out 2Wiki Cells

The Bonferroni-robust four answer one question: which signals beat length? A sharper question is when length itself stops being informative. On the two held-out 2Wiki cells, raw answer length is at or below chance for predicting retrieval need: `tokens_generated` reaches AUROC **0.457** on Qwen × 2Wiki and **0.518** on Gemma × 2Wiki. The verbosity-compensation pattern Zhang et al. (2024) document on the same LongBench bundle does not transfer to `lift_em`: longer multi-hop traces are no longer reliably retrieval-helpful. The four Bonferroni-robust survivors clear the length baseline by a wider margin on these cells than on HotpotQA, meaning the raw-AUROC protocol miscounts the length signal on exactly the cells where retrieval matters most.

4.3 FCT Discriminates Strongest on the Held-Out MoE Cell

The cells where length fails are the cells where FCT succeeds. On Gemma × 2Wiki, the cell with the worst length proxy (`tokens_generated` 0.518), FCT reaches raw AUROC **0.784**, a $+0.266$ absolute gap over length (Table 1). The same pattern repeats on Qwen × 2Wiki: where the length proxy is below chance (0.457), FCT's margin ($+0.147$)

is wider than on either HotpotQA cell. The answer-trajectory primitive carries the most independent information where the length proxy collapses.

Cell	n	helped	FCT AUROC
Qwen × HotpotQA	200	31	0.677
Qwen × 2Wiki	200	34	0.604
Gemma × HotpotQA	200	19	0.642
Gemma × 2Wiki	200	24	**0.784**

Table 1: Per-cell raw AUROC of FCT on `lift_em`=+1.

4.4 Survivor Set Is Robust to a Semantic-Equivalence Judge

The natural alternative explanation for the survivor matrix is that it reflects EM's string-matching artifact rather than a length-orthogonal signal. To rule this out we relabelled both arms of every paired trace via a GPT-4o-Mini judge (Janiak et al.'s exact choice) and re-ran the audit, addressing the correctness-function critique of Janiak et al. (2025) and Ielanskyi et al. (2025). EM and the judge agree at Cohen's $\kappa = 0.707$ (substantial agreement; Landis and Koch, 1977). Eight of the nine paired-significant survivors under EM remain significant under the judge, with a single swap (`etc` loses, `flare.max` gains) that leaves the Bonferroni-robust four intact. FCT itself *strengthens* under the judge from $\Delta = +0.190$ to $\Delta = +0.222$, the opposite of what a string-level EM artifact would predict under semantic relabeling (App. C).

4.5 The K=5 Multi-Sample Variant Is Null

The paired-bootstrap CI on $\Delta_{K=5-K=1}$ for `flip_rate` straddles zero on both architectures (Qwen -0.002, $[-0.040, +0.031]$; Gemma $+0.033$, $[-0.025, +0.092]$; App. H). The audit reports K=5 throughout because that was the capture protocol when the ablation returned null; for deployment K=1 delivers the same triggering performance at 5× less compute per boundary.

5 Discussion

5.1 Why Length Predicts Retrieval-Helpfulness

On the secondary `closed_em`=0 target, `tokens_generated` reaches raw AUROC 0.539 on the headline pool, competitive with several internal signals. This pattern matches Zhang et al. (2024)'s verbosity-compensation mechanism. A plausible causal chain is length → verbosity compensation → uncertainty → incorrectness → retrieval need, and length predicts `closed_em`=0 moderately well because it proxies the upstream incorrectness step. The paired-oracle target `lift_em`= + 1 strips that step by conditioning on closed-book wrongness, which is why length collapses on the held-out 2Wiki cells (§4.2).

5.2 Why Answer-Trajectory Stability Predicts Retrieval Need

The mechanism behind FCT reads naturally as *premature commitment*, the interpretation Datta et al. (2026) give to their `AnswerSwitches` and $t^\star$ primitives. The model decides on an answer early and continues to verbalize reasoning that supports the commitment, with most subsequent steps as post-decision tail. Datta et al. apply this observation to early stopping: if the model has decided, no further compute is needed. Retrieval triggering reads the same primitive in the opposite causal direction. The traces where the model has not decided, where the answer trajectory continues to flip through the reasoning, are exactly the traces where external evidence would resolve the uncertainty. The paired-oracle target `lift_em`= + 1 formalizes that question directly. The same primitive supports two complementary operations: detect stabilization to *stop*, detect non-stabilization to *retrieve*.

6 Conclusion

We presented the first length-residualized audit of the dynamic-RAG trigger pool, extending the protocol of Vashurin et al. (2025) to a paired-oracle target (`lift_em`= + 1). On two open-weight chat models over HotpotQA and 2WikiMultiHopQA, nine signals beat the length baseline at 95% paired-bootstrap and four survived Bonferroni at k=15 across three families. The strongest length-orthogonal trigger is **FCT**, our forward-looking variant of the answer-trajectory primitive, exceeding length by $\Delta = +0.190$ AUROC and reaching raw 0.784 on the held-out Gemma × 2Wiki cell. The survivor set is robust to a GPT-4o-Mini semantic-equivalence judge ($\kappa = 0.707$), with FCT itself strengthening under the judge. The same answer-trajectory primitive supports two complementary deployment operations: detect stabilization to *stop*, detect non-stabilization to *retrieve*.

Limitations

Signal-level audit, not end-to-end pipelines. We audit trigger *signals* as defined in their original papers, not full deployment pipelines (threshold tuning, retriever choice, prompt format). Our claims pertain to signal-level discriminative power on `lift_em`.

Bonferroni knife-edge. Two of the four Bonferroni-robust survivors (`ent.max`, `min_majority`) carry $\sim$99.67% lower bounds within 5×10^{-3} of zero (App. A). The four-survivor count is therefore fragile to small perturbations. The FCT and DRAGIN-RIND lower bounds are robust at $+0.047$ and $+0.039$.

Architecture and dataset coverage. The headline pool covers two architectures and two multi-hop QA benchmarks. The cheapest extensions are a third open-weight chat model (e.g., Llama-3.1-8B-Instruct) and a third multi-hop benchmark (MuSiQue, Trivedi et al., 2022). The current FCT Bonferroni lower bound ($+0.047$) is the statistical floor additional cells would have to displace.

Dataset framing: held-out, not OOD. We treat 2WikiMultiHopQA as a *held-out* cell relative to HotpotQA, not as out-of-distribution in the strict sense: both are multi-hop QA over Wikipedia with comparable question complexity. The cell-dependent length behavior (§4.2) is a dataset-specific finding on multi-hop QA rather than a claim about OOD generalization, and a stricter cross-distribution evaluation is future work.

Correctness function. We use exact-match equality on both arms of the paired-oracle target. The paired design (§3.1) and the LLM-as-Judge robustness check (App. C) partially insulate the headline against this choice. Containment-match and F_1 variants remain future work.

Non-thinking-mode chat models. The audited models are non-thinking-mode chat variants. Datta et al. (2026) developed their answer-trajectory primitive on thinking-mode reasoning models with explicit `</think>` tokens. Our $K{=}5$ boundary-completion adaptation is not byte-faithful to that protocol.

Risks. Our work concerns evaluation methodology for retrieval triggering on open-weight chat models and public multi-hop QA benchmarks. We release no new datasets or trained models, and direct risks are limited. One methodological risk is that the length-residualized protocol could be cited in isolation to defend deployed dynamic-RAG systems whose triggers track length more strongly than residualization alone would reveal. We therefore recommend reporting raw, length-residualized, and LLM-judge-validated AUROC together (§4.4) rather than substituting one for another.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *arXiv preprint*. ArXiv:2310.11511 [cs].

Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. 2024. INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection. *arXiv preprint*. ArXiv:2402.03744 [cs].

Jinkun Chen, Fengxiang Cheng, Sijia Han, and Vlado Keselj. 2026. "I May Not Have Articulated Myself Clearly": Diagnosing Dynamic Instability in LLM Reasoning at Inference Time. *arXiv preprint*. ArXiv:2602.02863 [cs.AI].

Ayan Datta, Zhixue Zhao, Bhuvanesh Verma, Radhika Mamidi, Mounika Marreddy, and Alexander Mehler. 2026. Large Language Models Decide Early and Explain Later. *arXiv preprint*. ArXiv:2604.22266 [cs.CL].

Dongxin Guo, Jikun Wu, and Siu Ming Yiu. 2026. When to Retrieve During Reasoning: Adaptive Retrieval for Large Reasoning Models. *arXiv preprint*. Version Number: 1.

Hanghui Guo, Jia Zhu, Shimin Di, Weijie Shi, Zhangze Chen, and Jiajie Xu. 2025. DioR: Adaptive Cognitive Detection and Contextual Retrieval Optimization for Dynamic Retrieval-Augmented Generation. *arXiv preprint*. Version Number: 2.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing A Multi-hop QA Dataset for Comprehensive Evaluation of Reasoning Steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Mykyta Ielanskyi, Kajetan Schweighofer, Lukas Aichberger, and Sepp Hochreiter. 2025. Addressing Pitfalls in the Evaluation of Uncertainty Estimation Methods for Natural Language Generation. *arXiv preprint*. ArXiv:2510.02279 [cs.LG].

Denis Janiak, Jakub Binkowski, Albert Sawczyn, Bogdan Gabrys, Ravid Shwartz-Ziv, and Tomasz Jan

Kajdanowicz. 2025. The Illusion of Progress: Re-evaluating Hallucination Detection in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 34716–34733, Suzhou, China. Association for Computational Linguistics.

Zhengbao Jiang, Frank F. Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. Active Retrieval Augmented Generation. *arXiv preprint*. ArXiv:2305.06983 [cs].

J. R. Landis and G. G. Koch. 1977. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2021. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *arXiv preprint*. ArXiv:2005.11401 [cs.CL].

Bo Li, Tian Tian, Zhenghua Xu, Hao Cheng, Shikun Zhang, and Wei Ye. 2026. Modeling Uncertainty Trends for Timely Retrieval in Dynamic RAG. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(37):31527–31535.

Haolang Lu, Minghui Pan, Ripeng Li, Guoshun Nan, Jialin Zhuang, Zijie Zhao, Zhongxiang Sun, Kun Wang, and Yang Liu. 2026. Streaming Hallucination Detection in Long Chain-of-Thought Reasoning. *arXiv preprint*. ArXiv:2601.02170 [cs.AI].

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822, Toronto, Canada. Association for Computational Linguistics.

Zhenjiang Mao, Anirudhh Venkat, Artem Bisliouk, Akshat Kothiyal, Sindhura Kumbakonam Subramanian, Saithej Singhu, and Ivan Ruchkin. 2026. Confidence over Time: Confidence Calibration with Temporal Logic for Large Language Model Reasoning. *arXiv preprint*. ArXiv:2601.13387 [cs.CL].

Satwik Pandey, Suresh Raghu, and Shashwat Pandey. 2026. SELFDOUBT: Uncertainty Quantification for Reasoning LLMs via the Hedge-to-Verify Ratio. *arXiv preprint*. ArXiv:2604.06389 [cs].

Andrea Santilli, Adam Golinski, Michael Kirchhof, Federico Danieli, Arno Blaas, Miao Xiong, Luca Zappella, and Sinead Williamson. 2025. Revisiting Uncertainty Quantification Evaluation in Language Models: Spurious Interactions with Response Length Bias Results. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 743–759. ArXiv:2504.13677 [cs.CL].

Zhihong Shao, Yeyun Gong, Yelong Shen, Minlie Huang, Nan Duan, and Weizhu Chen. 2023. Enhancing Retrieval-Augmented Large Language Models with Iterative Retrieval-Generation Synergy. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9248–9274, Singapore. Association for Computational Linguistics.

Weihang Su, Yichen Tang, Qingyao Ai, Zhijing Wu, and Yiqun Liu. 2024. DRAGIN: Dynamic Retrieval Augmented Generation based on the Information Needs of Large Language Models. *arXiv preprint*. Version Number: 3.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. MuSiQue: Multi-hop Questions via Single-hop Question Composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10014–10037, Toronto, Canada. Association for Computational Linguistics.

Roman Vashurin, Maiya Goloburda, Preslav Nakov, and Maxim Panov. 2025. UNCERTAINTY-LINE: Length-Invariant Estimation of Uncertainty for Large Language Models. *arXiv preprint*. ArXiv:2505.19060 [cs.CL].

Zezhong Wang, Xingshan Zeng, Weiwen Liu, Yufei Wang, Liangyou Li, Yasheng Wang, Lifeng Shang, Xin Jiang, Qun Liu, and Kam-Fai Wong. 2025. Chain-of-Probe: Examining the Necessity and Accuracy of CoT Step-by-Step. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 2586–2606, Albuquerque, New Mexico. Association for Computational Linguistics.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, Brussels, Belgium. Association for Computational Linguistics.

Zijun Yao, Weijian Qi, Liangming Pan, Shulin Cao, Linmei Hu, Liu Weichuan, Lei Hou, and Juanzi Li. 2025. SeaKR: Self-aware Knowledge Retrieval for Adaptive Retrieval Augmented Generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 27022–27043, Vienna, Austria. Association for Computational Linguistics.

Jianxiong Zhang, Bing Guo, Yuming Jiang, Haobo Wang, Bo An, and Sean Du. 2026a. Harnessing Reasoning Trajectories for Hallucination Detection via

Answer-agreement Representation Shaping. *arXiv preprint*. ArXiv:2601.17467 [cs].

Long Zhang, Wei-neng Chen, Feng-feng Wei, and Zi-bo Qin. 2026b. When Does a Language Model Commit? A Finite-Answer Theory of Pre-Verbalization Commitment. *arXiv preprint*. ArXiv:2605.06723 [cs.AI].

Yusen Zhang, Sarkar Snigdha Sarathi Das, and Rui Zhang. 2024. Verbosity $\neq$ Veracity: Demystify Verbosity Compensation Behavior of Large Language Models. *arXiv preprint*. ArXiv:2411.07858 [cs.CL].

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *arXiv preprint*. ArXiv:2306.05685 [cs.CL].

A Bonferroni-Corrected Survivor Set

Table 2 reports the Bonferroni-corrected ∼99.67% paired-bootstrap CI of $\Delta = \mathrm{AUROC}(s_{\text{resid}}) - \mathrm{AUROC}(\texttt{tokens_generated})$ at $k{=}15$ ($\alpha = 0.05/15 = 0.0033$) on the headline pool. Bonferroni k is the number of signals tested for the survivor-matrix claim. Four signals retain a strictly positive lower CI bound under correction: FCT, DRAGIN-RIND, `min_majority`, and `ent.max`.[1]

Signal	Δ	95% CI	99.67% CI (Bonf.)
Bonferroni-robust:			
FCT	+0.190	[+0.093, +0.287]	[+0.047, +0.329]
DRAGIN-RIND	+0.167	[+0.082, +0.249]	[+0.039, +0.286]
min_majority	+0.149	[+0.054, +0.240]	[+0.004, +0.284]
ent.max	+0.137	[+0.050, +0.223]	[+0.001, +0.263]
Nominal-95% only:			
CABV	+0.137	[+0.041, +0.229]	[−0.006, +0.273]
flip_rate $K{=}5$	+0.135	[+0.037, +0.233]	[−0.012, +0.277]
u_cons.max	+0.120	[+0.028, +0.213]	[−0.017, +0.257]
flip_rate $K{=}1$	+0.116	[+0.016, +0.217]	[−0.030, +0.261]
etc ($\Delta^2 H$)	+0.107	[+0.018, +0.196]	[−0.026, +0.235]

Table 2: Bonferroni-corrected survivor set on the headline pool (Qwen+Gemma, $n{=}800$ paired, `lift_em`$= +1$, $B{=}10{,}000$). The `etc` ($\Delta^2 H$) signal is the Entropy-Trend Constraint trigger of Li et al. (2026).

[1] $k{=}15$ was the multiple-comparison count fixed at test time, when the audited pool included a CABV×LAAS composite that we later dropped from presentation (§3.3). The current table reports 16 audited signals plus the length baseline. The Bonferroni-corrected CI half-widths in the table (≈ 0.12–0.14 AUROC at $B{=}10{,}000$) dominate any plausible k-sensitivity at this signal pool size. We report the test as run.

B Full Headline Survivor Matrix

Table 3 reports all 16 audited signals on the headline pool with length-residualized AUROC and Δ vs. `tokens_generated`.

C Correctness-Function Robustness: LLM-as-Judge

Protocol. For each of the $n{=}800$ paired traces in the headline pool, we re-labelled both arms via an OpenAI `gpt-4o-mini` semantic-equivalence judge under a strict three-sentence system prompt: "You are a strict semantic-equivalence judge for multi-hop QA. Your task is to decide whether a model-predicted answer is semantically equivalent to a gold answer. Reply with exactly one word: YES or NO. No explanation." The judge target is

$$\texttt{lift_em}_{\text{JUDGE}}(q) = \mathbb{1}\big[\text{judge}(a_o^{(q)}){=}\text{YES} \wedge \text{judge}(a_c^{(q)}){=}\text{NO}\big]. \quad (4)$$

We re-ran the full headline survivor matrix with paired bootstrap $B{=}10{,}000$.

Inter-label agreement. EM identifies 108 helped traces and the judge identifies 113; the intersection is 87. The two labels jointly produce a multi-class Cohen's $\kappa = 0.707$ on the 3-class `lift_em` $\in \{+1, 0, -1\}$ label set, in the substantial-agreement band of Landis and Koch (1977).

Survivor matrix transfer. Table 4 reports the headline survivor matrix under both correctness functions side-by-side. Eight of the nine paired-significant survivors under EM remain significant under the judge, with a single swap: `etc` loses significance and `flare.max` gains it. FCT strengthens under the judge from $\Delta = +0.190$ [+0.093, +0.287] to $\Delta = +0.222$ [+0.132, +0.312] (+17% in point estimate, +42% in 95% CI lower-bound margin), the opposite of what a string-level EM artifact would predict. All four Bonferroni-robust survivors retain nominal-95% significance under the judge.

Caveat. **?** document that GPT-4-tier judges exhibit a verbosity bias, preferring longer responses; our cross-judge check demonstrates *pattern stability* rather than full verbosity-independence. The load-bearing length-orthogonality argument rests on the UNCERTAINTY-LINE residualization (§3.2) and the paired-oracle design (§3.1); the judge check confirms the survivor matrix is not an artifact of any single correctness function.

Signal	Family	Resid AUROC	Δ vs. tokens	95% CI
`tokens_generated`	length	0.531	—	(baseline)
FCT	answer-trajectory	0.691	+0.190	[+0.093, +0.287]
DRAGIN-RIND	attention	0.640	+0.167	[+0.082, +0.249]
min_majority	answer-trajectory	0.626	+0.149	[+0.054, +0.240]
ent.max	token-probability	0.611	+0.137	[+0.050, +0.223]
CABV	answer-trajectory	0.641	+0.137	[+0.041, +0.229]
flip_rate K=5	answer-trajectory	0.640	+0.135	[+0.037, +0.233]
u_cons.max	K=3 text	0.593	+0.120	[+0.028, +0.213]
flip_rate K=1	answer-trajectory	0.621	+0.116	[+0.016, +0.217]
etc ($\Delta^2 H$)	entropy-trend	0.580	+0.107	[+0.018, +0.196]
late_flips	answer-trajectory	0.586	+0.081	[−0.016, +0.175]
flare.max	token-probability	0.553	+0.080	[−0.008, +0.165]
hvr.max (SELFDOUBT-T1) (Pandey et al., 2026)	hedging-marker	0.505	+0.032	[−0.089, +0.150]
hidden_norm.range	hidden (geom.)	0.499	+0.026	[−0.058, +0.108]
drift_cosine.max	hidden (geom.)	0.493	+0.020	[−0.064, +0.101]
seakr_k5.logdet	K=5 hidden	0.491	+0.018	[−0.077, +0.112]
var_l2_window.max	hidden (geom.)	0.473	+0.000	[−0.091, +0.090]

Table 3: Full cross-architecture survivor matrix on the headline pool (Qwen2.5-7B + Gemma-4-26B-A4B-it × {HotpotQA, 2WikiMHQA}, n=800 paired traces). All AUROCs are length-residualized (§3.2). 95% CIs are paired-bootstrap (B=10,000) of $\Delta = \mathrm{AUROC}(s_{\text{resid}}) - \mathrm{AUROC}(\texttt{tokens_generated})$. **Bold**: Bonferroni-robust at k=15 (lower bound of the 99.67% CI strictly positive; see Table 2).

Signal	Δ_{EM}	Δ_{JUDGE}
FCT	+0.190	**+0.222**
DRAGIN-RIND	+0.167	+0.157
min_majority	+0.149	+0.155
ent.max	+0.137	+0.125
CABV	+0.137	+0.177
flip_rate K=5	+0.135	+0.178
u_cons.max	+0.120	+0.157
flip_rate K=1	+0.116	+0.171
etc ($\Delta^2 H$)	+0.107	+0.088*
flare.max	+0.080*	+0.115

Table 4: Side-by-side Δ under EM and LLM-as-Judge on the headline pool (n=800, B=10,000). *95% CI straddles zero. FCT *strengthens* under the judge. Eight of nine paired-significant EM survivors are preserved, and all four Bonferroni-robust survivors retain nominal-95% significance.

D Uncertainty-Line Reference Replication

To pre-empt the reimplementation critique, we ran our protocol (Eq. 2) and the Vashurin et al. (2025) reference implementation side-by-side on the same trace cohort, decomposing the comparison into two ε components. $\varepsilon_{\text{protocol}}$ isolates the math difference (our $\log(\texttt{tokens}+1)$ single-cohort OLS against the reference MinMax-scaled linear regression on the same cohort); $\varepsilon_{\text{sampling}}$ isolates train/test-split sampling noise at n=200 (Table 5).

Arch	$\varepsilon^{\text{prot}}_{\max}$	$\varepsilon^{\text{prot}}_{\text{mean}}$	$\varepsilon^{\text{prot}}_{\text{med}}$	$\varepsilon^{\text{samp}}_{\max}$
Qwen2.5	0.029	0.012	0.012	0.111
Gemma	0.027	0.0095	0.0047	0.106

Table 5: $\varepsilon_{\text{protocol}} = |\text{ours}_{\text{resid}} - \text{LINE}_{\text{full}}|$; $\varepsilon_{\text{sampling}} = |\text{LINE}_{\text{full}} - \text{LINE}_{\text{split}}|$ (latter averaged over 3 seeds). Both architectures show $\varepsilon_{\text{protocol}} \le 0.029$, well below the conventional threshold of 0.05. Train/test sampling noise ($\varepsilon_{\text{sampling}}$) dominates the protocol gap by a factor of 3–4×, indicating that the $\log(\texttt{tokens}+1)$ vs. raw-MinMax math difference is small relative to evaluation-set variance at n=200.

E FCT vs. Concurrent Commitment-Time Primitives: Operational Differences

FCT (§3.3) is motivated by the final-switch index $t^\star = \max\{i : A_i \neq A_{i+1}\}$ of Datta et al. (2026), with a concurrent forward-looking commitment-time primitive proposed by Zhang et al. (2026b) on the model's finite-answer continuation distribution. The three are operationally distinct on five axes (Table 6).

F CABV: Continuous Answer-Belief Velocity

CABV is a continuous between-step generalization of `flip_rate`: at each populated step we build a Laplace-smoothed empirical distribution over the trace's K=5 answers and average the Jensen–Shannon divergence between consecutive

Axis	Datta $t^\star$	Zhang τ_{commit}	FCT (ours)
Temporal direction	Backward (last switch)	Forward (first stable)	Forward (first stable)
Signal class	Per-step answer	Continuation log-odds	K=5 majority answer
Sampling regime	K=1 forced-completion	Single deterministic state	K=5 stochastic
Application target	Early stopping	Pre-verbalization commitment	Retrieval triggering
Trace model class	Thinking (`</think>`)	Delayed-verdict templates	Non-thinking chat

Table 6: Operational differences across three concurrent commitment-time primitives. Datta et al. (2026)'s $t^\star$ is backward-looking on per-step intermediate answers, applied to early stopping. Zhang et al. (2026b)'s τ_{commit} is forward-looking on the model's finite-answer continuation distribution (log-odds in binary tasks), applied to pre-verbalization commitment analysis. FCT (ours) is forward-looking on K=5-majority answer extracts, applied to retrieval triggering. For monotonically converging trajectories, $t_{\text{commit}} \approx t^\star + 1$; under K=5 majority sampling on non-thinking-mode chat models this equivalence does not hold in general.

steps. CABV reduces to `flip_rate` when each step concentrates on a single answer, and is closest in spirit to the inference-time instability signal of Chen et al. (2026), defined on JSD between token distributions rather than K=5 answer distributions.

G Length-vs-Signal Spearman ρ per Cell

Table 7 reports Spearman ρ between `tokens_generated` and each signal per (arch $\times$ dataset) cell on closed-book traces, the length-confound diagnostic of Santilli et al. (2025) (their Table 2). $|\rho| \geq 0.5$ indicates a strong length confound on the signal.

H K=5 vs. K=1 Ablation

Table 8 reports the paired-bootstrap $\Delta_{K=5-K=1}$ for `flip_rate` on `lift_em`$= +1$, pooled per architecture. Both CIs straddle zero (§4.5).

I Adapter Notes

We audit signal-level discriminative power on `lift_em` (§6). Several primitives in the published trigger pool are originally framed as online per-token triggers or per-trace scores, and we evaluate their underlying statistic at the trace level. The per-signal notes below state the original primitive, our trace-level adaptation, and any sign or granularity drift.

ETC (Li et al., 2026). The original ETC operates on a token-level entropy stream; we evaluate $\Delta^2 H$ on the per-step mean entropy series captured at boundary-detection events (§3.3). We score $\max|\Delta^2 H|$ rather than the signed second-difference, so the adapter is magnitude-faithful but not sign-faithful.

Signal	Q×Hot	Q×Wiki	G×Hot	G×Wiki
DRAGIN-RIND	+0.40	+0.35	+0.38	+0.38
ent.max	+0.31	+0.26	+0.42	+0.32
etc ($\Delta^2 H$)	+0.32	+0.36	+0.46	+0.40
flare.max	+0.39	+0.35	+0.45	+0.32
u_cons.max	+0.40	+0.31	+0.37	+0.39
seakr_k5	+0.37	+0.44	+0.46	+0.43
drift_cosine.max	+0.20	+0.18	−0.07	−0.01
var_l2_window.max	+0.19	+0.23	+0.22	−0.00
hidden_norm.range	+0.24	+0.29	+0.36	+0.16
hvr.max	n/a*	+0.06	+0.30	**+0.51**
flip_rate K=5	+0.22	+0.12	+0.21	+0.35
flip_rate K=1	+0.17	+0.15	+0.27	+0.21
late_flips	+0.29	+0.37	+0.25	+0.30
min_majority	+0.18	+0.12	+0.14	+0.09
CABV	+0.22	+0.17	+0.22	+0.36
FCT	+0.20	+0.14	+0.15	+0.30

Table 7: Spearman ρ(`tokens_generated`, signal) per cell on closed-book traces. Bold: $|\rho| \geq 0.5$. The peer-pool signals (DRAGIN, ent.max, etc, flare.max, u_cons.max, seakr_k5) carry $\rho \in [+0.26, +0.46]$, `hvr.max` reaches $+0.51$ on Gemma×2Wiki, and the auxiliary hidden-state geometry diagnostics span $\rho \in [-0.07, +0.36]$. The answer-trajectory aggregations (bottom block) average ≈ 0.15 lower in $|\rho|$ than the peer-pool block, the per-cell quantitative version of the family-level finding from §4.1. *Qwen×Hot `hvr.max` is constant, so Spearman is undefined.

Arch	n	helped	$\Delta_{K=5-K=1}$	95% CI
Qwen	305	47	−0.002	$[-0.040, +0.031]$
Gemma	394	42	+0.033	$[-0.025, +0.092]$

Table 8: K=5 vs. K=1 `flip_rate` ablation on `lift_em`$= +1$, pooled per architecture.

FLARE (Jiang et al., 2023). The original FLARE is an online per-token retrieval trigger ($\max_t p_t < \tau$); we audit the underlying primitive (low-confidence detection) as a trace-level summary max over steps of $1 - \min p_t$. The trigger *policy* is not audited; the trigger *statistic* is.

SeaKR (Yao et al., 2025). We use K=5 EOS hidden states for compute efficiency, smaller than the paper's reported best-k interval $K \in [10, 25]$. The regularized Gram determinant we compute is the same primitive as the paper's self-aware uncertainty score. INSIDE (Chen et al., 2024) is the closest peer, using the EigenScore on the centered sample covariance; the K nonzero eigenvalues of our uncentered Gram and INSIDE's centered covariance are rank-equivalent up to centering.

SELFDOUBT-T1 (Pandey et al., 2026). HVR $= h/(v{+}1)$ is computed per step and aggregated as $\max$ over steps; the original paper computes a single trace-level HVR with a Tier-1 "HVR=0" high-precision correctness gate. The per-step adapter is recoverable as the paper's gate by setting our threshold to 0.

Concurrent commitment-time primitives. Operational differences between Datta et al. (2026)'s $t^\star$, Zhang et al. (2026b)'s τ_{commit}, and our FCT are tabulated in Appendix E.

J Reproducibility

All trace JSONLs are committed at `runs/v2_headline_{arch}/` for arch $\in$ {qwen2.5-7b-instruct, gemma-4-26b-a4b-it}. Slice manifests with SHA-256 checksums are at `runs/v2_headline_{arch}/checkpoints/slice_manifest.json`, and decoding command lines are at `runs/v2_headline_{arch}/logs/00_master.log`. Analysis scripts are at `analysis/r1_paired_audit.py`, `analysis/a1_k1_vs_k5_ablation.py`, `analysis/uncertainty_line_audit.py`, `analysis/moonshot_explore.py`, `analysis/llm_judge_audit.py`, and `analysis/e7_bonferroni_and_spearman.py` (the last produces Table 2 and Table 7). The judge cache contains 1,600 per-call JSONs (oracle and closed arm for each of 800 paired traces) at `analysis/llm_judge_cache/`. Compute box: single NVIDIA A800 80 GB, wall-clock $\sim$15 h.